

Schema Reference

Every table and column in the VinHack database, generated directly from the live schema. Derived columns are marked — those are filled by the database itself and must not be written by a client.

SQLite 3.45.1 9 tables · 83 columns 5 views · 4 triggers Generated 19 Sep 2026

students *One row per student.* 9 cols · 3 rows

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
student_id	INTEGER	PK	
name	TEXT	NOT NULL	
email	TEXT	NOT NULL UNIQUE	
semester	INTEGER		semester BETWEEN 1 AND 12
programme	TEXT		
registration_no	TEXT		
sleep_goal_minutes	INTEGER	NOT NULL	default 480
daily_study_goal_hours	REAL	NOT NULL	default 4.0
created_at	TEXT	NOT NULL	default datetime('now', 'localtime')

TABLE CONSTRAINTS & INDEXES

- CHECK sleep_goal_minutes BETWEEN 240 AND 720
- CHECK daily_study_goal_hours BETWEEN 0 AND 16

sleep_logs *One row per student per night.* 8 cols · 42 rows

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
sleep_id	INTEGER	PK	
student_id	INTEGER	FK → students.student_id NOT NULL	
sleep_date	TEXT	NOT NULL	
bedtime	TEXT		
wake_time	TEXT		
duration_minutes	INTEGER		duration_minutes BETWEEN 0 AND 1440 <i>Derived by trigger from bedtime and wake_time when not supplied.</i>
quality	INTEGER		quality BETWEEN 1 AND 5
created_at	TEXT	NOT NULL	default datetime('now', 'localtime')

TABLE CONSTRAINTS & INDEXES

- UNIQUE (student_id, sleep_date)
- CHECK wake_time IS NULL OR bedtime IS NULL OR wake_time > bedtime
- INDEX sleep_logs_student_date_idx on (student_id, sleep_date)

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
screen_id	INTEGER	PK	
student_id	INTEGER	FK → students.student_id NOT NULL	
date	TEXT	NOT NULL	
social_media_minutes	INTEGER	NOT NULL	default 0 social_media_minutes >= 0
entertainment_minutes	INTEGER	NOT NULL	default 0 entertainment_minutes >= 0
education_minutes	INTEGER	NOT NULL	default 0 education_minutes >= 0
communication_minutes	INTEGER	NOT NULL	default 0 communication_minutes >= 0
other_minutes	INTEGER	NOT NULL	default 0 other_minutes >= 0
total_screen_minutes	INTEGER	GENERATED	Generated column: the sum of the five category columns. Never inserted.

TABLE CONSTRAINTS & INDEXES

- UNIQUE (student_id, date)
- INDEX screen_time_student_date_idx on (student_id, date)

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
task_id	INTEGER	PK	
student_id	INTEGER	FK → students.student_id NOT NULL	
task_name	TEXT	NOT NULL	
subject	TEXT		
task_type	TEXT	NOT NULL	default 'other'
due_date	TEXT		
estimated_effort_hours	REAL		estimated_effort_hours >= 0
priority	TEXT	NOT NULL	default 'medium'
status	TEXT	NOT NULL	default 'pending'
completed_at	TEXT		Managed by trigger: set when status becomes 'completed', cleared otherwise.
created_at	TEXT	NOT NULL	default datetime('now', 'localtime')

TABLE CONSTRAINTS & INDEXES

- CHECK task_type IN ('assignment','exam','quiz','project','reading','lab','other')
- CHECK priority IN ('low','medium','high','urgent')
- CHECK status IN ('pending','in_progress','completed','cancelled')
- INDEX academic_tasks_student_due_idx on (student_id, due_date)
- INDEX academic_tasks_student_status_idx on (student_id, status)

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
event_id	INTEGER	PK	
student_id	INTEGER	FK → students.student_id NOT NULL	
event_name	TEXT	NOT NULL	
event_type	TEXT	NOT NULL	default 'other'
start_time	TEXT	NOT NULL	
end_time	TEXT	NOT NULL	
is_fixed	INTEGER	NOT NULL	default 1 is_fixed IN (0, 1)
location	TEXT		

TABLE CONSTRAINTS & INDEXES

- CHECK event_type IN ('class','exam','lab','meeting','personal','commute','other')
- CHECK end_time > start_time
- INDEX calendar_events_student_start_idx on (student_id, start_time)

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
mood_id	INTEGER	PK	
student_id	INTEGER	FK → students.student_id NOT NULL	
date	TEXT	NOT NULL	
mood_score	INTEGER		mood_score BETWEEN 1 AND 10
energy_score	INTEGER		energy_score BETWEEN 1 AND 10

TABLE CONSTRAINTS & INDEXES

- UNIQUE (student_id, date)
- INDEX mood_energy_student_date_idx on (student_id, date)

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
session_id	INTEGER	PK	
student_id	INTEGER	FK → students.student_id NOT NULL	
task_id	INTEGER	FK → academic_tasks.task_id	ON DELETE SET NULL, so logged effort survives the task being deleted.
start_time	TEXT	NOT NULL	
end_time	TEXT		
duration_minutes	INTEGER		duration_minutes >= 0 Derived by trigger from start_time and end_time when not supplied.
focus_rating	INTEGER		focus_rating BETWEEN 1 AND 5

TABLE CONSTRAINTS & INDEXES

- CHECK end_time IS NULL OR end_time > start_time
- INDEX study_sessions_task_idx on (task_id)
- INDEX study_sessions_student_start_idx on (student_id, start_time)

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
assessment_id	INTEGER	PK	
student_id	INTEGER	FK → students.student_id NOT NULL	
task_id	INTEGER	FK → academic_tasks.task_id	ON DELETE SET NULL, so a grade survives the task it came from being deleted.
subject	TEXT	NOT NULL	
title	TEXT	NOT NULL	
category	TEXT	NOT NULL	default 'other'
assessed_on	TEXT	NOT NULL	
score	REAL	NOT NULL	score >= 0
max_score	REAL	NOT NULL	default 100 max_score > 0
weight_percent	REAL		weight_percent BETWEEN 0 AND 100 Share of the final subject grade. Items without one are averaged evenly.
class_average	REAL		class_average >= 0
created_at	TEXT	NOT NULL	default datetime('now', 'localtime')

TABLE CONSTRAINTS & INDEXES

- CHECK category IN ('exam', 'quiz', 'lab', 'project', 'assignment', 'participation', 'other')
- CHECK score <= max_score
- INDEX assessments_student_subject_idx on (student_id, subject)
- INDEX assessments_student_date_idx on (student_id, assessed_on)

COLUMN	TYPE	KEYS & FLAGS	DEFAULTS & CONSTRAINTS
metric_id	INTEGER	PK	
student_id	INTEGER	FK → students.student_id NOT NULL	
date	TEXT	NOT NULL	
sleep_duration	INTEGER		sleep_duration BETWEEN 0 AND 1440
sleep_quality	INTEGER		sleep_quality BETWEEN 1 AND 5
social_media_minutes	INTEGER		social_media_minutes >= 0
total_screen_minutes	INTEGER		total_screen_minutes >= 0
pending_tasks	INTEGER		pending_tasks >= 0
high_priority_tasks	INTEGER		high_priority_tasks >= 0
hours_until_nearest_deadline	REAL		
available_study_hours	REAL		available_study_hours >= 0
study_hours_completed	REAL		study_hours_completed >= 0
energy_score	INTEGER		energy_score BETWEEN 1 AND 10
productivity_score	REAL		productivity_score BETWEEN 0 AND 100

TABLE CONSTRAINTS & INDEXES

- UNIQUE (student_id, date)
- INDEX daily_metrics_student_date_idx on (student_id, date)

views

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VIEW	COLUMNS
v_assessment_pct	assessment_id, student_id, task_id, subject, title, category, assessed_on, score, max_score, weight_percent, class_average, created_at, percent, class_delta
v_committed_by_day	student_id, date, committed_hours
v_open_tasks	task_id, student_id, task_name, subject, task_type, due_date, estimated_effort_hours, priority, status, completed_at, created_at, hours_until_due, is_overdue
v_study_by_day	student_id, date, study_hours, avg_focus, session_count
v_subject_grades	student_id, subject, items, weighted_percent, mean_percent, weight_recorded, avg_class_delta, last_assessed_on

triggers

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TRIGGER	ON TABLE
academic_tasks_completed_ins	academic_tasks
academic_tasks_completed_upd	academic_tasks
sleep_logs_duration_ins	sleep_logs
study_sessions_duration_ins	study_sessions

Conventions. Dates are ISO-8601 text YYYY-MM-DD; timestamps are YYYY-MM-DD HH:MM:SS in UTC. Enumerated values are TEXT with a CHECK constraint. Booleans are INTEGER 0/1. Foreign keys are off by default in SQLite and the pragma is per-connection, so all access goes through `vinhack.db.connect()`.

Row counts reflect the development seed, not production data.